

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2014

H1 BIOLOGY
8875

Paper 2

2 Hours

Additional materials:
Writing papers / OMS

Date / Day: Tuesday 19th August 2014**INSTRUCTIONS TO CANDIDATES**

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

Section A	
1	/8
2	/9
3	/8
4	/8
5	/7
Section B	
6	/20
7	/20
TOTAL	/60

This question paper consists of 14 printed pages

Question 1

Porins are transmembrane proteins found on the outer membrane of mitochondria. Their function is to transport substances across the membrane. **Fig. 1.1a** and **Fig. 1.1b** show the side view and top view of the structure of a single polypeptide of porin respectively.

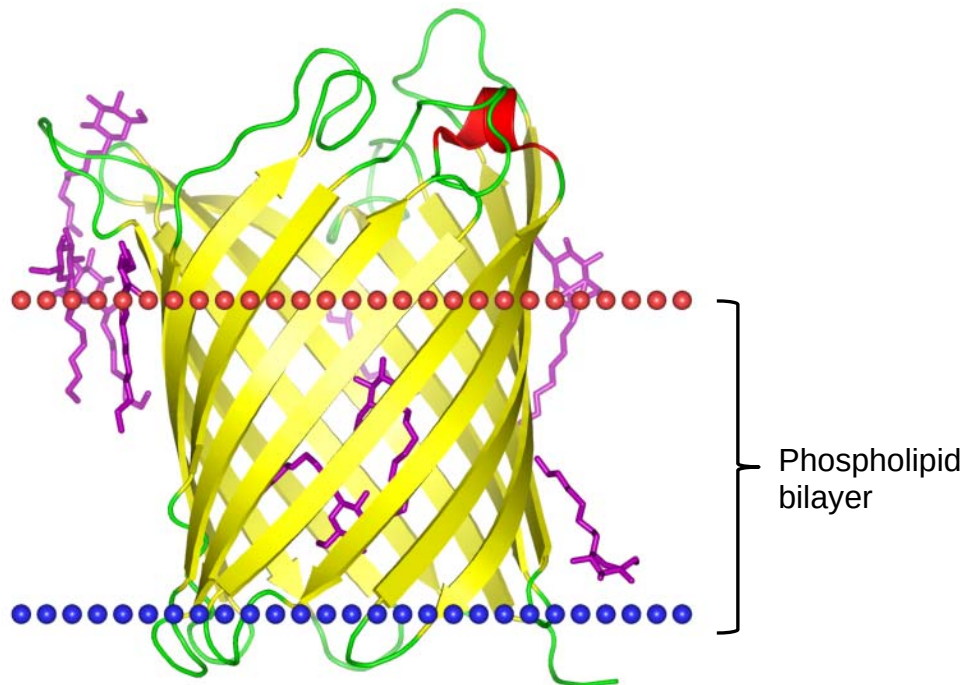


Fig. 1.1a



Fig. 1.1b

(a) Name a covalent bond present within the protein porin. [1]

1. Peptide bond/disulfide bond

(b) With reference to **Fig. 1.1a** and **Fig. 1.1b**,

(i) Name a non-covalent bond present within the protein porin. [1]

1. Hydrogen bond/ Ionic bond

(Reject hydrophobic interactions as not considered a bond)

(ii) Describe the structure of the protein porin. [2]

1. Contains **several anti-parallel β -pleated sheets**;

2. **Folding of polypeptide chain to form specific 3D conformation/tertiary structure.**

A researcher intends to study the transport of substances carried out by porins across the outer mitochondrial membrane. The researcher begins by isolating mitochondria from rat liver cells and placing them in 5 different solutions, each containing a different substance with a concentration of 10 arbitrary units as shown in **Table 1.2**.

Table 1.2

Solution	Content of Solution
1	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions
4	10 AUs of protein insulin
5	10 AUs of acidic amino acid glutamate

The concentrations of the substances in the intermembrane space of the mitochondria were measured before the experiment and after 15 minutes of being placed in the solutions. The results obtained by the researcher are shown in **Table 1.3** on the following page (page 4).

Table 1.3

Solution	Content of Solution	Concentration of substances in mitochondrial intermembrane space before experiment	Concentration of substances in mitochondrial intermembrane space after 15 minutes
1	10 AUs of Na ⁺ ions	1 AU of Na ⁺ ions	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions	2 AUs of Cl ⁻ ions	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions	2 AUs of H ⁺ ions	10 AUs of H ⁺ ions
4	10 AUs of protein insulin	0 AU of protein insulin	0 AU of protein insulin
5	10 AUs of acidic amino acid glutamate	0 AU of acidic amino acid glutamate	9 AUs of acidic amino acid glutamate

(c) With reference to **Table 1.2** and **Table 1.3**, suggest how the outer mitochondrial membrane is responsible for the concentrations of substances in mitochondrial intermembrane space after 15 minutes. [4]

1. **Phospholipid bilayer / hydrophobic core** (of cell surface membrane) is **impermeable to ions**;
2. **transport proteins/porins** are embedded across the membrane;
3. to allow **facilitated diffusion of ions/ small molecules, example glutamate**;
4. **insulin is too large** to be transported across the membrane.

[Total: 8 marks]

Question 2

Scientists often employ the method of fluorescence imaging to study the process of nuclear division in cells. This method involves the use of fluorescence tagged antibodies to label specific structures within the cell, so that their movements and behaviours can be followed over the course of nuclear division.

A group of SR scientists decided to employ the aforementioned method to study the changes in centromere number in an animal cell undergoing meiosis. Initial testing showed that each centromere could be visualised as a single dot and therefore the total number of dots can be taken to correspond to the total number of centromeres within the cell. **Table 2.1** shows the number of dots and hence centromeres observed per cell, at various stages of meiosis.

stage of Meiosis	Prophase I	Prophase II	Anaphase II	Telophase II
Number of dots observed per cell	16			16

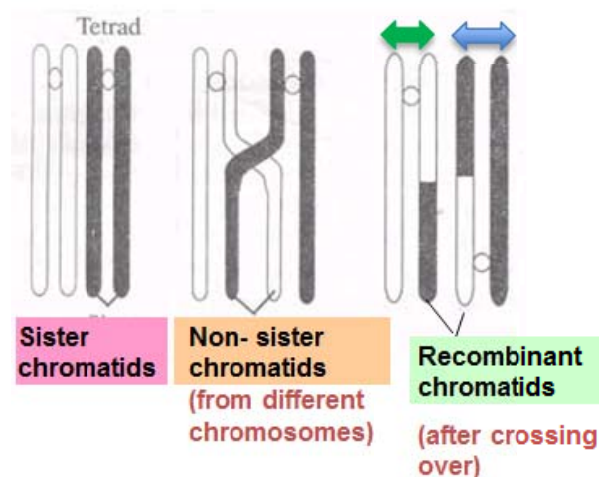
With reference to **Table 2.1**,

Predict the number of dots (i.e. centromeres) observed in prophase II and anaphase II respectively. [2]

1. Prophase II: 8;
2. Anaphase II: 16

Explain your answer to part (a) (i). [2]

1. Prophase II: Homologous chromosomes are separated equally into 2 daughter cells/ haploid set of chromosomes is formed at the end of meiosis I/ following cytokinesis;
2. Anaphase II: Centromeres divide in anaphase II, thus doubling number of centromeres.



When the same experiment was conducted while the cell was in interphase, prior to meiosis I, no distinct dots could be observed in the nucleus. Instead, the entire nucleus emitted fluorescence.

(a) Suggest a reason for this observation. [1]

1. Chromatin had not condensed into discrete chromosomes with centromeres/ OWTTE.

Apart from using fluorescence imaging to determine the changes in centromere numbers, this technique was also employed by the SR scientists to study the movement of chromosomes during meiosis in the animal cell. More specifically, the changes in distance between 2 corresponding regions of non-sister chromatids of a pair of homologous chromosomes (**X**) were measured over the course of the first meiotic division. The results are shown in **Fig 2.1** below.

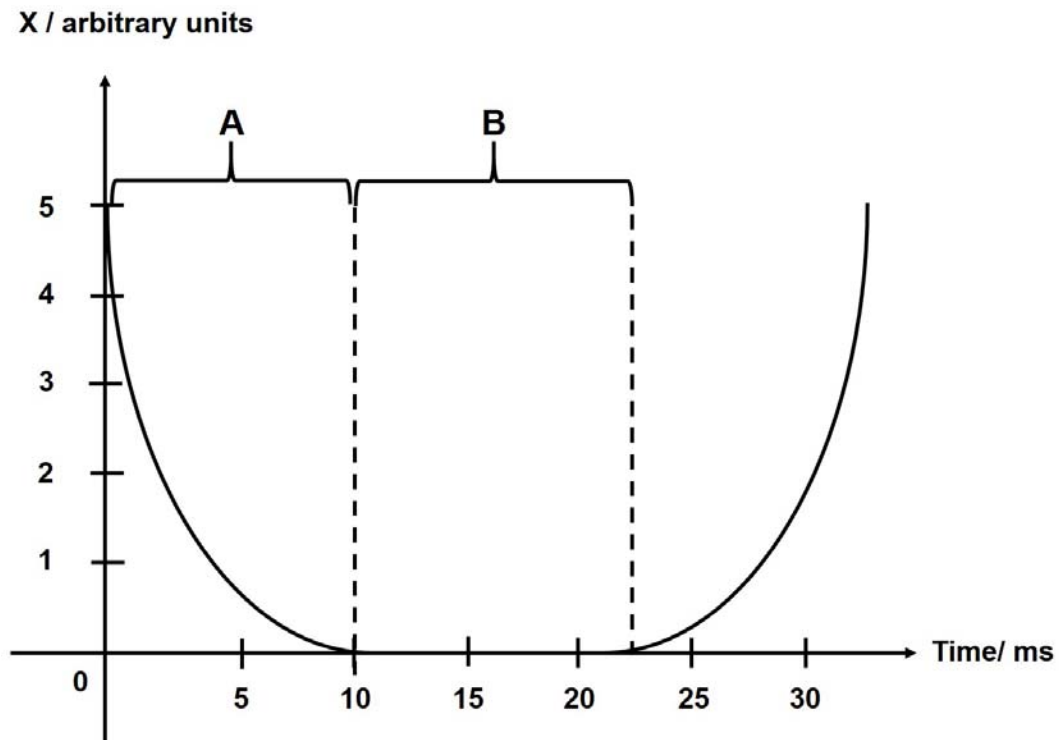


Fig 2.1

(b) with reference to **Fig 2.1**, explain the changes in **X** for

(i) Region A. [2]

1. X decreases from 5 to 0 arbitrary units;
2. Homologous chromosomes pair up via synapsis to form bivalents

(ii) Region B. [2]

1. X remains constant at 0 arbitrary units;
2. Chiasma formation and crossing over occurs between non-sister chromatids of homologous chromosome

[Total: 9 marks]

Question 3

In an experiment to study ATP synthesis in plants, chloroplasts are isolated, and the pH levels in the various compartments are monitored. The graph below shows the results of the experiment.

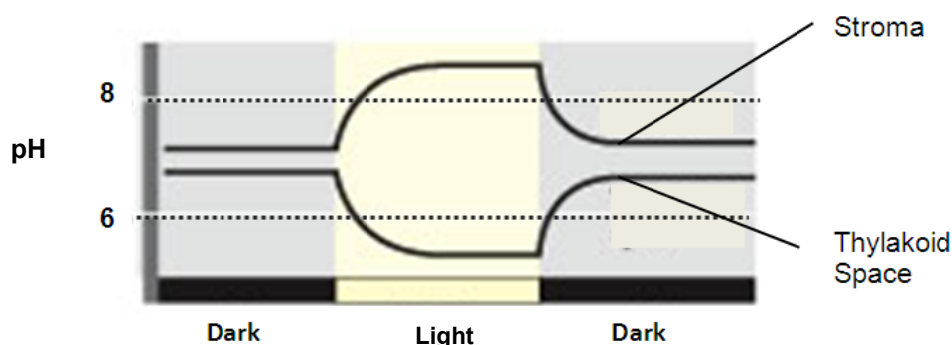


Fig 3.1

(a) Explain the changes in pH seen in **Fig 3.1** upon illumination. [3]

1. Illumination of chloroplasts causes the excitation of the electrons from the reaction centre.
2. This initiates the electrons flow down the ETC, energy lost is used to pump H^+ from the stroma into the thylakoid.
3. Hence the increase in pH in the stroma and the decrease in pH (increase in H^+) in the thylakoid space

(b) Describe a similarity and a difference in the production of ATP in chloroplasts and in mitochondria. [2]

Similarity:

1. Flow of electron down the electron transport chain of progressively increasing electronegativity.
2. Generation of proton motive force / electrochemical / H^+ gradient
3. Diffusion of H^+ from through ATP synthase complex, energy used for the phosphorylation of ADP to ATP (ref to chemiosmosis, but R: just chemiosmosis without description)

Difference:

	Photophosphorylation	Oxidative phosphorylation
Electron donor	Reaction centre chlorophyll a / water	Reduced coenzyme, NADH/FADH
Final electron acceptor	NADP	Molecular <u>Oxygen</u>
Source of energy for phosphorylation	Light	Oxidation of glucose / NADH / FADH (to be discussed)
*Direction of H^+ pumped	Pumped from stroma to thylakoid space	Pumped from matrix to intermembranal space
*Direction of H^+ diffuse	from thylakoid space to stroma	from intermembranal space to matrix

* not the best to quote

Fig 3.2 shows some of the reactions during the light-independent reactions of photosynthesis.

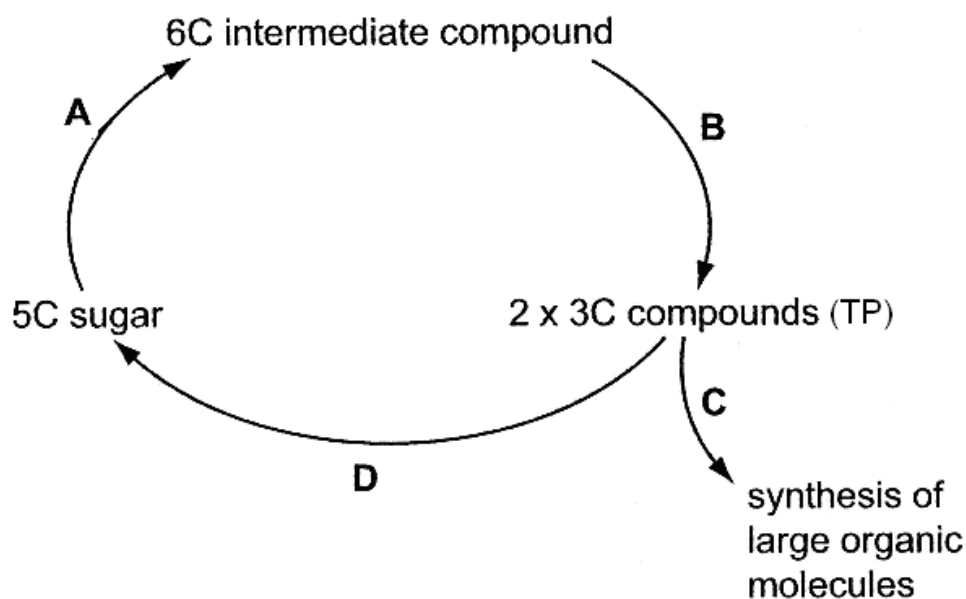


Fig 3.2

(c) With reference to **Fig 3.2**,

- (i) In which stage(s) will both products of the light-dependent reactions be used? [1]

1. **B**

- (ii) Describe how the products of the light-dependent reactions are used in the light-independent stage of photosynthesis. [2]

1. **ATP is used to regenerate ribulose biphosphate from triose phosphate (TP).**
2. **NADPH is used to reduce glycerate biphosphate (accept alternative name) to form triose phosphate**

[Total: 8 marks]

Question 4

Adalia bipunctata, the 2-spot ladybug (or ladybird), is a small beetle which exists in two forms. The typical form has red wing cases with two black spots. The red colour is a signal for predators of its unpleasant taste. The other less common form, melanic, has black wing case with two red spots. **Fig 4.1** shows the 2 different forms of ladybugs.

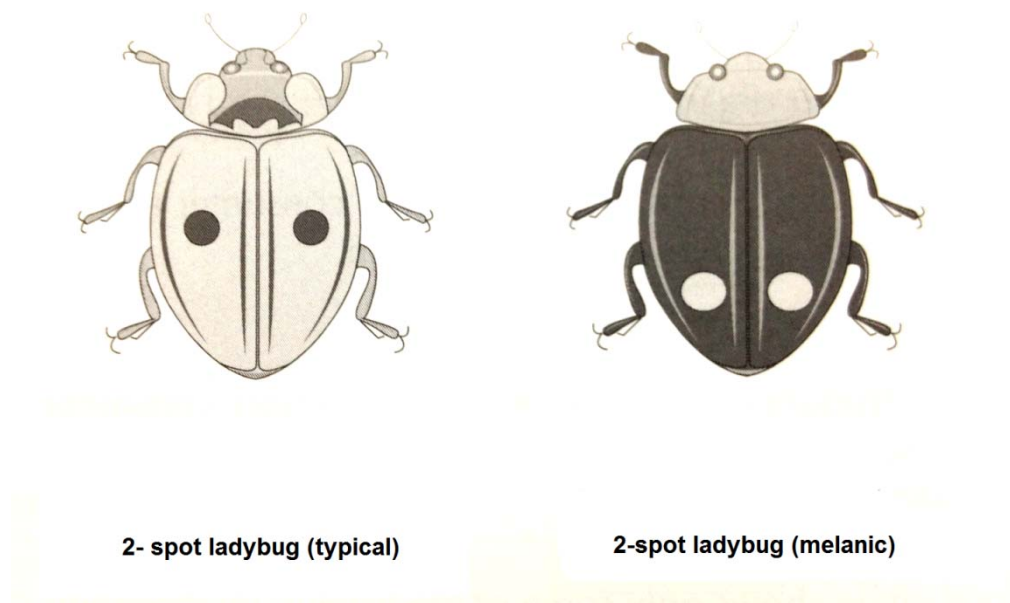


Fig 4.1

The melanic form used to be very common in industrial areas of Britain but declined after 1960. **Fig 4.2** shows the frequency of melanic ladybugs over the years and the correlation to smoke present in the air.

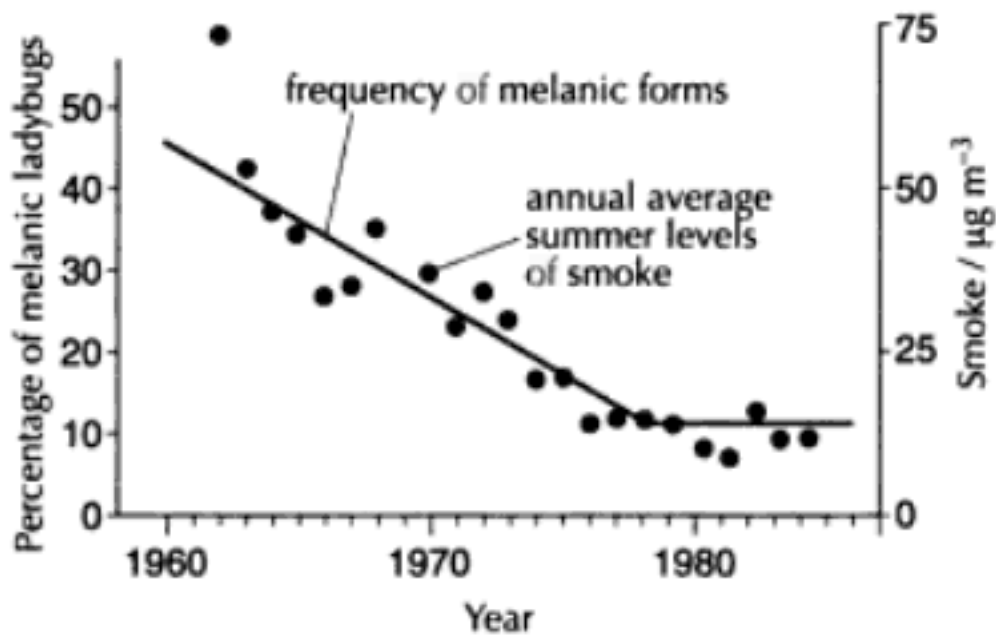


Fig 4.2

(a) Using **Fig 4.2**, explain how the process of natural selection had led to the changes in the frequencies of the melanic ladybug over the years. [4]

1. There is pre-existing genetic variations among the ladybugs- typical and melanic;
2. The percentage of melanic ladybugs decreased from 45% to 10% when amount of smoke produced decreased from 60 to 15 μgm^{-3} ; (data)
3. When there is a change in environment, i.e. decreased in production of smoke, thus the selection pressure changes;
With decreased in smoke in the air, the melanic form is unable to camouflage itself from its predator and was at a selective disadvantage (selected against); typical form selected for as it is not eaten by predators due to its red colour signalling unpleasant taste.
4. The typical form will survive to maturity, reproduce and pass on favourable alleles to its viable offspring leading to more of the typical form in the population

(b) A region of DNA coding for NADH dehydrogenase subunit was isolated for analysis between the three species - *Adalia bipunctata* (2-spot ladybug), *Coccinella septempunctata* (7-spot ladybird) and *Propylea quatuordecimpunctata* (14-spot ladybird). Part of the DNA sequence coding for NADH dehydrogenase subunit found in these three species are shown in **Fig. 4.3**.

<i>Adalia bipunctata</i>	ACCAGCCTGTGCATCGATGACGACTAAGTGATACCATAAAGACT
<i>Coccinella septempunctata</i>	ACTAGCCTGTGCATCGATGACGACTAAGTGATACCATAAAGACT
<i>Propylea quatuordecimpunctata</i>	ACGAGCCTGTGCATCGATGACGACTAAGTGATACCATAAAGACT

Fig. 4.3

(i) Suggest why NADH dehydrogenase gene is used for analysis. [1]

1. NADH dehydrogenase is an important enzyme in all organisms thus it will be found in every organism compared.

(ii) What is the term used to describe the relationship between the sequences shown in **Fig 4.3**? [1]

1. Homology

Note: This kind of homology is called molecular homology; not genetic homology/DNA homology etc.

(iii) Explain how the relationship between the DNA sequences coding for NADH dehydrogenase subunit in **Fig. 4.3** provides evidence to support the theory of evolution. [2]

1. Comparing homologous DNA sequences for NADH dehydrogenase subunit between species; DNA sequences for NADH dehydrogenase subunit are slightly different / only one base difference in *Propylea quatuordecimpunctata* compared to the other two species;
2. This shows that the different species might have descended with modifications from a common ancestor;

[Total: 8 marks]

Question 5

A monkey homozygous for a mutation on the insulin gene suffers from Type 1 diabetes. An experiment utilizing genetic engineering techniques was conducted to obtain genetically modified stem cells in order to treat the disease.

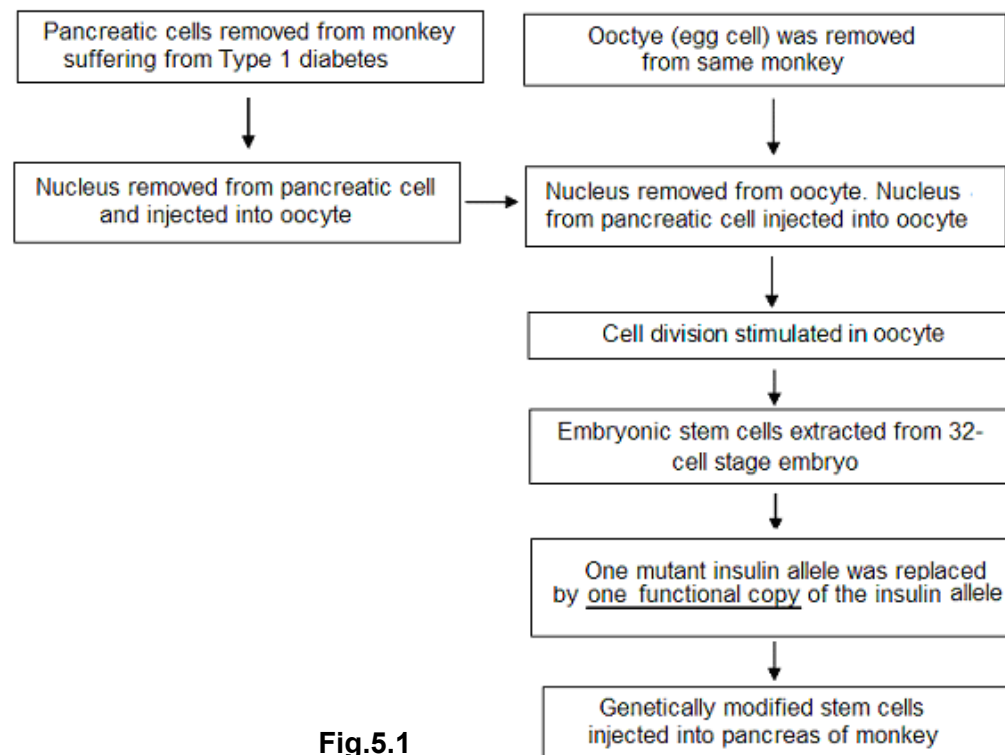


Fig.5.1

(a) Describe the role of embryonic stem cells in a normally developing embryo. [2]

1. Can differentiate / specialize due to differential gene expression / different sets of genes activated;
2. Develop into the three layers: ectoderm, mesoderm, endoderm;
3. Further differentiate into all tissues and organs of the organism

(b) Compare between embryonic stem cells and adult stem cells. [2]

Similarity (Any 1):

1. are capable of long term-self-renewal/ undergo numerous cycles of cell division/mitosis, while maintaining undifferentiated;
2. are capable of differentiating into specialized cells types under appropriate conditions,

Difference:

1. adult stem cells were multipotent/ giving rise to a limited range of cell types, whilst embryonic stem are pluripotent/ giving rise to almost any type of cell.

(c) Suggest why it is sufficient to insert a single functional allele into the stem cell. [1]

1. Functional insulin allele is dominant

(d) Discuss **one** ethical concern regarding the use of embryonic stem cells. [1]

1. Destruction of blastocyst
2. Donation of excess fertilised eggs from IVF clinic requires informed consent

[Total: 7 marks]

Section B (20 Marks)

Question 6

(a) Explain how the structure of DNA differs from tRNA. [4]

(1 mark for each pair of contrasting statements)

	DNA	tRNA
Basic Unit	<u>DNA nucleotide/ deoxyribonucleotide</u>	<u>RNA nucleotide/ ribnonucleotide</u>
Pentose sugar component	<u>Deoxyribose</u> sugar	<u>Ribose</u> sugar
Nitrogenous bases	Adenine (A), guanine (G), cytosine (C) and thymine	Adenine (A), guanine (G), cytosine (C) and uracil
Structure	Always a <u>double-stranded helix</u> , except in some viruses, DNA may be single-stranded.	Almost always occurs as a <u>single-stranded</u> molecule.

(b) Describe the polymerase chain reaction and contrast it with DNA replication. [10]

Describe (max 5);

1. PCR is a *in vitro* technique to amplify specific sequence of DNA in a short period of time;
2. Place the microcentrifuge/eppendorf tube in an automated thermocycler;
3. 3 step process: Denaturation + Annealing + Extension;
4. Denaturation: Heat the DNA mixture to about 94 - 96°C for 30 sec; to separate the double-stranded DNA to form single stranded DNA; by disrupting the hydrogen bonds between complementary bases;
5. Annealing: Cool the DNA mixture to about 50 - 65°C for 30 sec; allowing complementary DNA primers to anneal to the 3' ends of both single-stranded DNA template;
6. Extension: Heat the DNA mixture by increasing the temperature to about 70-72°C for 1.5 min;
7. To allow for Taq polymerase to bind to the DNA template where primer has annealed;
To add nucleotides to 3' OH ends of primers by complementary base pairing;
8. using the single strand DNA (target seq) as a template catalyses formation of phosphodiester bonds between adjacent nucleotides;
9. The cycle is repeated 30-40 times;
the number of DNA molecules is doubled every cycle, producing multiple copies of amplified target sequence.

Differences: (1 mark for each pair of contrasting statements)

	PCR	DNA Replication
Type of primer	DNA primer used	RNA primer required
Replication segment	Only specific segment/sequences of DNA molecule replicated	Entire DNA molecule replicated
Strand separation	High temperatures at 95° C	Enzyme <u>helicase</u> unwind DNA double helix
Synthesis of DNA strands	Both daughter stands are synthesized continuously	One daughter strand is synthesized continuously ; while the other daughter strand is synthesized discontinuously as short Okazaki fragments
Enzyme and property	<u>Taq polymerase</u> is used which does not have proofreading ability	<u>DNA polymerase</u> is used which has proofreading ability
Use of DNA ligase	DNA ligase is not required as both daughter stands are synthesized continuously	DNA ligase is required to catalyse the formation of phosphodiester bonds between Okazaki fragments to form a single DNA strand
Removal of RNA primer	No need to remove primer used as it is DNA	Additional step to remove RNA primer and replace it with DNA

(c) Explain, with an example, how a change in the sequence of the DNA nucleotide may affect the amino acid sequence in a protein and hence the phenotype of the organism. [6]

1. Example: **sickle – cell anaemia** (award once only)
2. **Single base / nucleotide substitution** within the gene coding for β -globin chain, with **thymine** substitutes with **adenine**;
3. a **codon in mRNA** is **changed** from **GAA to GUA**, thus results in a **change of amino acid** with a **different R group**;
4. **Hydrophilic/polar glutamic acid** is replaced by **hydrophobic/non-polar valine**;
5. **different R groups interactions** are formed, leading to changes in **folding of polypeptide**, thus, **changes its tertiary structure in haemoglobin S (HbS)**;
6. **At low oxygen concentrations**, the **hydrophobic regions on different HbS molecules would stick or aggregate together**, causing HbS molecules to **polymerise**, forming **rigid rod-like fibres**; this cause **red blood cells to change from circular, biconcave shape to sickle-shape**;
7. **Sickle-shaped red blood cells may obstruct blood capillaries, hence, (multiple) organs deprived of oxygen/resulting in organ damage**
These cells (**more rigid and fragile than normal red blood cells**) **haemolyse readily, leading to phenotype of sickle cell anaemia.**

Question 7

(a) Describe the main structural features and functions of endoplasmic reticulum, nucleolus and ribosome. [10]

	Structure	Function
Endoplasmic reticulum (Max 5)	1. comprised of an extensive network of <u>membrane bound tubules and sacs</u> called <u>cisternae</u>; 2. which is continuous with the nuclear membrane; 3. <u>Rough ER</u>: cytosolic surface is embedded with <u>ribosomes</u>; 4. <u>Smooth ER</u> : more tubular (REJECT “flattened sacs”) & surface lacks ribosomes	Function of Rough ER 1. To package proteins into transport vesicles which bud off from the ER and transport the proteins to the Golgi apparatus; 2. Involved in initial glycosylation of proteins (addition of carbohydrate side chain) Function of Smooth ER (max 1) 1. Synthesis and transport of lipids e.g. membrane phospholipids/ steroid hormones; 2. Detoxification of drugs and poisons, especially in liver cells; 3. Storage of calcium ions/ regulation of calcium ion concentration in muscle cells -calcium ions are involved in muscle contraction; 4. Involved in carbohydrate metabolism (conversion of glucose-6-phosphate to glucose) in liver cells
Ribosome (Max 3)	1. comprised of a large subunit and a small subunit; 2. the subunits are comprised of ribosomal RNA and protein	1. Protein synthesis occurs at the ribosome; 2. where the translation of messenger mRNA takes place for the production of the primary structure/amino acid sequence of protein

Nucleolus Max 2	1. densely stained (granules and fibres); 2. rich in rRNA genes	1. Synthesis of rRNA/ transcription of rRNA genes ; 2. Assembly of rRNA and ribosomal proteins into large and small subunits
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(b) Explain how gel electrophoresis is used to analyse nucleic acids. [6]

1. DNA fragments mixed with loading dye are loaded into wells at one end of an agarose gel, near the negatively charged electrode;
2. A loading dye is added to the DNA samples, which allows the process of electrophoresis to be tracked / helps DNA samples to sink into the wells;
3. As DNA fragments are negatively charged due to the phosphate group, they will migrate towards the positive end of the gel (anode);
4. DNA fragments encounter resistance from the agarose gel when moving through it as the agarose gel forms a cross-linked matrix that functions as a 'molecular sieve';
5. Gel electrophoresis separates DNA fragments by size / molecular weights;
6. Smaller size fragments migrate more rapidly/at a faster rate than large fragments and located further away from the well because smaller size fragments have lower resistance in moving through the pores of the gel, thus the DNA is separated into bands;
7. A ladder containing fragments of known sizes/ can be used as a standard, thus allowing sizes of sample/unknown fragments to be estimated by visual comparison of the DNA bands;
8. The invisible DNA fragments can be stained/intercalated with ethidium bromide and UV light is used to visualize the DNA bands

- (b) Discuss the problems associated with the expression of eukaryotic genes in prokaryotes and suggest how these problems can be overcome.
[4]

	Problem	Solution
1.	Prokaryotes <u>cannot remove introns</u> and thus <u>cannot convert pre-mRNA to mRNA</u> (no post-transcriptional modifications). This is because bacterial cells do <u>not have RNA splicing machinery</u>, eg enzymes, spliceosomes.	Clone gene of interest as <u>cDNA</u>, thus avoid the additional steps to remove introns. Use <u>reverse transcription method</u>; <u>mRNA</u> is used as <u>template</u> and <u>reverse transcriptase</u> catalyses the synthesis of <u>complementary DNA</u>; followed by <u>DNA polymerase III</u> that catalyses synthesis of another cDNA, making it <u>double stranded</u>.
2.	<u>RNA polymerases in bacterial cells</u> may <u>not recognize eukaryotic promoter</u>, thus gene is not transcribed.	Use an <u>expression vector</u> containing <u>prokaryotic promoter</u>, just upstream of a restriction site where the eukaryotic gene can be inserted. Bacterial RNA polymerases recognise prokaryotic promoter and can express the gene downstream to it.
3.	Lack of <u>post-translational modifications of eukaryotic proteins</u> due to the <u>lack of enzymes</u>.	Use of <u>eukaryotic cell systems</u> like <u>yeasts</u> as <u>host cells</u>; Enable glycosylation etc, for <u>correct folding</u> into the <u>tertiary and quaternary structures</u> so that proteins produced are <u>functional</u>.